



Effects of STATCOM device on power systems STATCOM cihazının güç sistemleri üzerindeki etkileri

Haşim Koçakelçi^{1*}, Yurdağül Benteşen Yakut²

¹Department of Electric Electronic Engineering, Faculty of Engineering, Dicle University, Diyarbakir, Türkiye.
fourthauthor@e-mail.address

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Abstract

Voltage stability is very important in electrical systems. The voltage and stability of the system can be significantly impacted by reactive power flows. Static Synchronous Compensator (STATCOM) is one of the converter-based Flexible Alternating Current Transmission Systems (FACTS) devices used for reactive power compensation. In this study, a voltage regulation simulation was carried out via STATCOM in a 400 kV and 2*12 MVar power system located in Kiziltepe District of Mardin Province, which is a joint project of TUBITAK (The Scientific and Technological Research Council of Türkiye) Marmara Research Center Energy Institute and TEİAŞ (Turkish Electricity Transmission Corporation) 16th Regional Directorate. The bus voltage from the distribution bus of a power transmission system was controlled by the STATCOM despite the changes in the load. Firstly, the working principle of STATCOM was examined, then STATCOM was installed on the distribution bus of a power transmission system located in Kiziltepe District of Mardin Province and its simulation was performed in the MATLAB program. As a result of the simulation, it was observed that STATCOM keeps the busbar voltage at the desired values, provides fast reactive compensation against load changes and increases the voltage stability of the system.

Keywords: FACTS devices, Voltage stability, Reactive power compensation, STATCOM

Öz

Gerilim kararlılığı elektrik sistemlerinde çok önemlidir. Sistemin voltajı ve kararlılığı, reaktif güç akışları tarafından önemli ölçüde etkilenebilir. Statik Senkron Kompansatör (STATCOM), reaktif güç kompanzasyonu için kullanılan dönüştürücü tabanlı Esnek Alternatif Akım İletim Sistemleri (FACTS) cihazlarından biridir. Bu çalışmada, TÜBİTAK Marmara Araştırma Merkezi Enstitüsü ile TEİAŞ 16. Bölge Müdürlüğü'nün ortak projesi olan, Mardin İli Kiziltepe İlçesi'nde bulunan 400 kV ve 2*12 MVar gücündeki bir enerji sisteminde STATCOM aracılığıyla gerilim düzenleme simülasyonu gerçekleştirilmiştir. Bir enerji iletim sisteminin dağıtım baralarından gelen bara voltajı, yükteki değişikliklere rağmen STATCOM tarafından kontrol edilmiştir. Öncelikle STATCOM'un çalışma prensibi incelenmiş, ardından STATCOM, Mardin İli Kiziltepe İlçesi'nde bulunan bir enerji iletim sisteminin dağıtım sistemine kurularak MATLAB programında simülasyonu gerçekleştirilmiştir. Simülasyon sonucunda, STATCOM'un bara voltajını istenen değerlerde tuttuğu, yük değişimlerine karşı hızlı reaktif kompanzasyon sağladığı ve sistemin gerilim kararlılığını artırdığı gözlemlenmiştir.

Anahtar kelimeler: FACTS cihazları, Gerilim kararlılığı, Reaktif güç kompanzasyonu, STATCOM

1 Introduction

Electrical energy generation, transmission, and distribution are all included in power systems. At the same time, power systems must be dependable, continuous, and efficient due to their extensive use. Power systems transmit electricity from power generation facilities to points of use via transmission lines and transformers. A significant engineering feat is needed in this process to transmit energy without loss and deliver it to the locations of consumption at the appropriate level and quality. The uninterrupted and dependable delivery of electrical energy to all consumers is guaranteed by careful power system design and management. Energy production in both new and current electricity production facilities must be provided to the customer in the most efficient manner, at the lowest cost, and with the least amount of loss, despite the growing demand for energy. Minimizing losses and operating and planning power systems in accordance with the age we live in are among the most important problems [1].

Because of the growing demand for energy, it is crucial to operate power systems in a region that is in close proximity to the system's stability limits. As the energy demand increases, it is crucial to balance the voltage instability, as the stability limit will result in a decrease in voltage stability. The power

system's equilibrium can be defined as the capacity to maintain the voltage amplitude on the load bus within specific confines under all circumstances. [2].

In recent years, the need for fast compensation in power systems has become increasingly important. Because they can satisfy the quick compensation requirement of transmission systems, power electronic components are favored in compensation application [3].

In the late 1980s, FACTS was presented to the world by EPRI (Electric Power Research Institute) in America [4]. The dynamic stability of the power system can be improved and voltage collapses can be avoided if the reactive compensation of transmission lines is completed promptly using thyristors and other semiconductor components, as has been known since the 1980s. The FACTS concept, introduced by EPRI in the late 1980s, is based on the replacement of mechanical controllers with safe and fast power electronic components. This change aims to ensure more efficient use of existing power system capacity and to increase controllability. FACTS devices replace traditional hardware with modern power electronics-based equipment to ensure power flow and voltage control [5]. The FACTS concept was introduced in 1988 by Dr. Narain

*Corresponding author/Yazışılan Yazar

Hingorani, Vice President of the Electrical Systems Division of EPRI.

FACTS, a system or static equipment based on power electronics that can be controlled and provides control of many AC (Alternative Current) transmission systems in order to increase power transfer, calls for the replacement of mechanical controllers with power electronic components that can operate safely and quickly, according to IEEE (Institute of Electrical and Electronics Engineer). This makes it feasible to facilitate control of the ever growing power system capacity and guarantee its most efficient use [4].

The main components of FACTS controller structures are power electronic elements. These components make it possible to quickly control electrical energy. These controllers, which can respond quickly, offer the opportunity to intervene at the desired level in short-term instabilities [6]. However, the control of electrical energy can be achieved quite quickly, depending on the working capacities of these elements. The most advanced of these controllers are STATCOM and SSSC (Static Synchronous Series Compensator) controllers. STATCOM is a parallel reactive power equipment that has the ability to produce and consume reactive power and to change the output values to control specific parameters of the electric power system. With the use of power electronic components, STATCOM can rapidly regulate the value and phase angle of three-phase voltages [7].

STATCOM is the general term used in the transmission system. In the distribution system, it is known as D-STATCOM [8]. Due to the widespread usage of semiconductor power components, many of the loads linked to distribution systems in an integrated way today exhibit inductive characteristics, and the distribution system experiences non-phase and non-linear currents. These factors lead to notable losses in the system's energy power quality [9]. While STATCOM plays a significant role in managing a power system's transmission and distribution networks in general, it has recently become increasingly crucial in managing renewable energy sources. By integrating STATCOM into the optimal power flow (HTOPF) algorithm in Hydro-Thermal structures, it can improve the voltage values of the system by injecting or withdrawing the reactive power required for the grid into the system, thus contributing to keeping the system under control [10]. STATCOM can also be used to do research on a mixed renewable energy generation system that combines fuel cells and solar panels [11]. The STATCOM device on the Matlab/Simulink program contained in this study can also be used to create simulations pertaining to the performance analysis of photovoltaic-wind hybrid energy systems [12]. Additionally, STATCOM can be made more effective in the aforementioned application areas. Numerous studies have been conducted on this topic, and the use of fuel cells from energy storage systems has improved STATCOM's efficacy on power systems [13].

STATCOM is a FACTS device that uses converters. Additionally, research indicates that it performs differently and more effectively in oscillation damping than SSSC, IPFC, and UPFC, all of which fall under the same category [14]. To put it briefly, STATCOM contributes significantly to industrial facilities, renewable energy sources, particularly photovoltaic-wind hybrid energy systems, and the generation, transmission, and distribution of electricity, among many other fields [15], [16], [17], [18], [19], [20].

A nation can advance technologically by developing power systems that are both high-quality and efficient. A nation enters the category of developed nations if its technological capabilities advance. When we examine the most technologically advanced nations in the world today, we find that these nations are always enhancing and developing new power systems. Because of this, every nation creates its own systems within the context of data analysis from power system research.

In this paper, discussed the transmission of power via a distribution center in the province of Mardin's Kızıltepe region. The data flow on the power system in question was investigated by installing a FACTS device (STATCOM) on the electrical transmission system. At the same time, the power system in question was replicated in a computer environment. It was investigated how the power system may be enhanced by contrasting the actual data with the data acquired following the simulation.

2 Facts devices

Energy losses occur in the transmission and distribution processes of electrical energy in power systems. These losses are usually caused by various factors such as resistance in conductors, magnetic losses in transformers, clearance losses due to gaps between overhead lines and cables, and other losses in distribution equipment. Energy losses can have significant economic and environmental consequences. Energy losses raise the cost of energy, which, economically speaking, decreases energy efficiency. Energy losses have the potential to increase greenhouse gas emissions and other pollutants into the atmosphere, as well as result in surplus energy output. As a result, lowering power system energy losses, improving energy efficiency, and minimizing environmental effects are all crucial objectives. In order to do this, technologies like FACTS and STATCOM are crucial parts of power systems that improve their dependability, effectiveness, and controllability. FACTS devices are static devices used to control voltage, current and power flow in power systems. These devices are typically implemented using thyristor-based switching technologies.

→ Thyristor Based Devices

- Thyristor Controlled Static Var Compensator (TCSVC)
- Thyristor Controlled Series Capacitor (TCSC)
- Thyristor Controlled Phase Angle Regulator (TCPAR)

→ Converter Based Devices

- Static Synchronous Series Compensator (SSSC)
- Unified Power Flow Controller (UPFC)
- Interline Power Flow Controller (IPFC)
- Static Synchronous Compensator (STATCOM)

In this study, STATCOM, one of the converter based devices, was examined.

3 Static synchronous compensator (STATCOM)

STATCOM is a type of FACTS device that is shunt-integrated into the power system and is used to control the voltage level and power factor in power systems. Also known as STATCOM or ASVC (Advanced Static Var Compensator) [21]. It is a very important technology that reduces the pressure on the voltage by improving the reactive power and significantly improves the power quality of the system [22].

If we define STATCOM in another way; STATCOM is a parallel connected VAR compensator. This static compensator is separate from the voltage of the alternating current system and can control capacitive or inductive output currents. Static synchronous compensator can be defined as static var compensator. That is, it is shunt connected. This type of flexible AC transmission control device has an output current that is inductive or capacitive and can be used independently [23].

3.1 Basic features of STATCOM

STATCOM is positioned on voltage source inverters. Based on the output parameters, three-phase balanced voltages can be obtained at grid frequency, with controllable amplitude and phase angle [24]. A basic analysis of the STATCOM structure reveals that it consists of one capacitor, three shunt (coupling) transformers, and a voltage source converter (VSC) [25].

Figure 1 shows the circuit diagram of STATCOM.

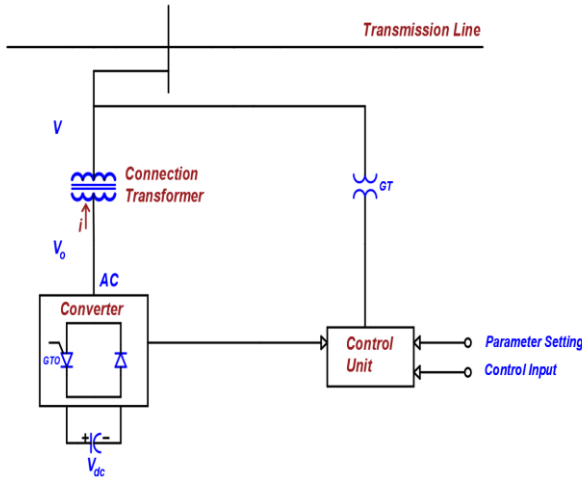


Figure 1. STATCOM circuit diagram.

The circuit elements of STATCOM are listed below.

- Voltage Source Converter (VSC)
- Coupling Transformer
- Control Unit
- DC Energy Source

3.2 STATCOM operating principles

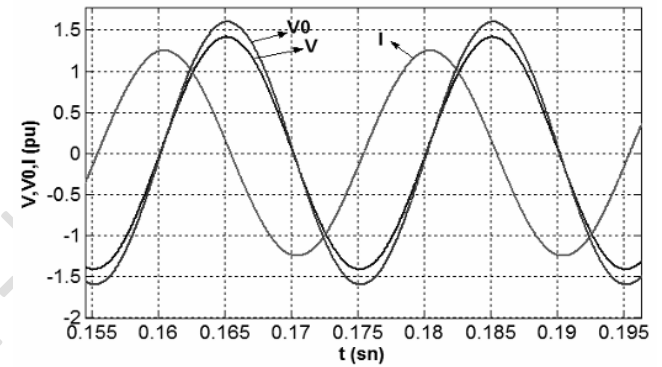
Most STATCOMs have suitable energy storage systems such as fuel cells, flywheel storage, supercapacitors, batteries, and other voltage-source converters with significant capacity for power modulation [26]. Serves as an alternative to other passive reactive power devices including capacitors and inductors or reactors. Variables have outputs in milliseconds and can supply and consume both capacitive and inductive variables. They are more suited for dynamic circumstances like fault conditions or unforeseen catastrophes due to their speed and capabilities. Suitable for voltage support and power factor adjustment.

By regulating the phase difference that may occur between the voltage value at the converter output and the voltage in the alternating current system; the effective energy transfer between the converter and the alternating current system can be regulated. The converter will demonstrate an efficient power flow from the DC energy source to the AC system if the converter voltage is maintained above the AC voltage. To put it

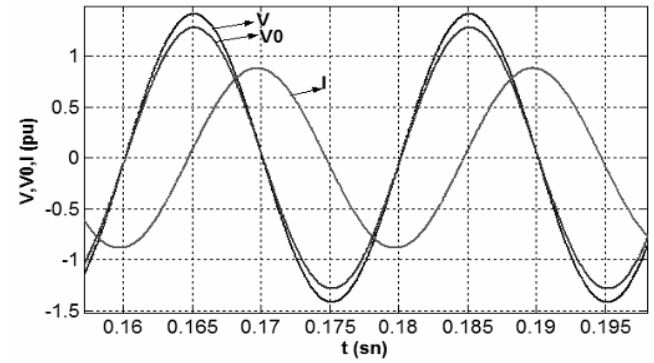
another way, the converter will effectively use power from the AC system if the output voltage is maintained behind the voltage of the AC system [21].

When the excess voltage value on the output side of the STATCOM exceeds the alternating current network voltage ($V_o > V$), an I current occurs in the AC network from the STATCOM through the reactance of the transformer and the device in question starts to produce reactive power in the power system. The converter's output voltage is in phase with the AC system voltage since STATCOM simply changes dormant power, and circuit losses are disregarded. Consequently, the line voltage is 90 degrees behind the current flowing on the converter's output side.

As a result, the gadget is in capacity mode. Figure 2.a illustrates this particular scenario. In the event that the STATCOM's output voltage is less than the AC system voltage ($V_o < V$), the system absorbs the reactive power of the network and a current is created from the alternating current system to the STATCOM device via the transformer's reactance [27], [28].



(a)



(b)

Figure 2. Reactive power behavior in STATCOM: (a) Supply of reactive power in capacitive case [21], (b) Dissipation of reactive power in the inductive case [21].

The converter current rating lags the line voltage by 90 degrees. (Energy loss is ignored) Thus, the device operates in induction mode. This situation is shown in figure 2.b when the voltage value at the converter output is equal to the voltage of the power transmission line ($V = V_o$), no reactive power transfer occurs between the alternating current system and the device [27],[28]. STATCOM produces inductive and capacitive currents within a certain range, separately from the alternating current system voltage. STATCOM can continue to provide

power flow to the system even if the system voltage drops to 0.15 pu (per unit) [21].

STATCOM has much better features than TCSVC which has the same function. Its response speed is much faster than TCSVC. Since it does not use capacitor or reactor groups to provide reactive power, it does not have the ability to create a resonance point. It takes comparatively less installation area since it doesn't employ reactor or capacitor groups and because semiconductor power devices with high capacity levels are present. The STATCOM can also be used to control the active power if it is connected to an appropriate energy storage device via a DC capacitor. The voltage-current characteristics of STATCOM and TCSVC devices are shown in figure 3. In low voltage networks, the current values provided by the STATCOM device to the system are at higher levels than the TCSVC device [21].

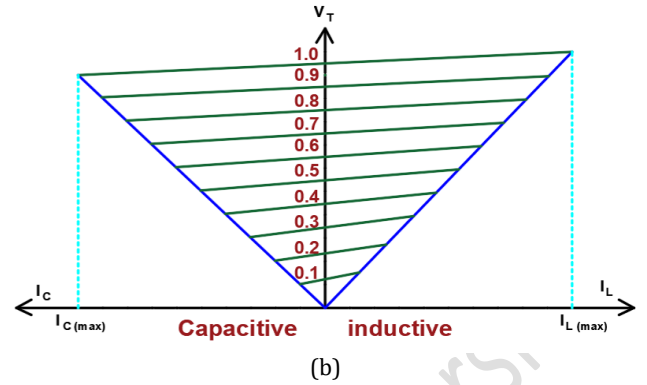
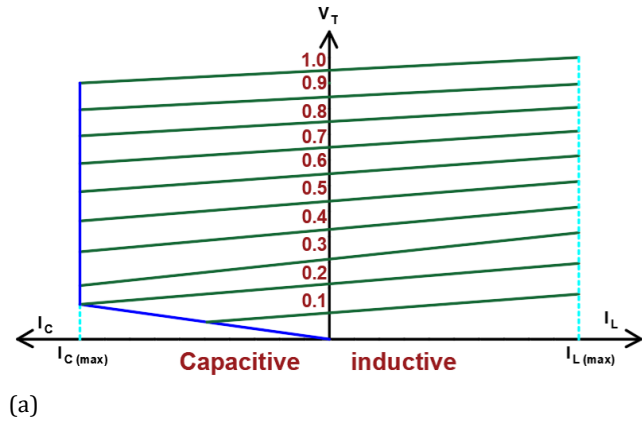


Figure 3. Current-Voltage characteristics: (a) STATCOM current voltage characteristic, (b) TCSVC current voltage characteristic.

STATCOM can provide full inductive or capacitive output currents at any system voltage. The reactive power compensation level injected into the grid by STATCOM is higher than TCSVC. This is due to the fact that reactive power in STATCOM falls in direct proportion to the voltage, whereas in TCSVC, it falls with the square of the voltage (at low voltage values). This suggests that STATCOM has far superior reactive power regulation compared to TCSVC [21].

3.3 STATCOM formulas and control parameters

When STATCOM is incorporated into a power system, reactive power can be added or removed. The STATCOM supplies reactive power to the circuit when the power system voltage is low, and it takes reactive power from the system when the voltage is high.

The mathematical formulas and control parameters of STATCOM are shown in a tabular form in Figure 4.

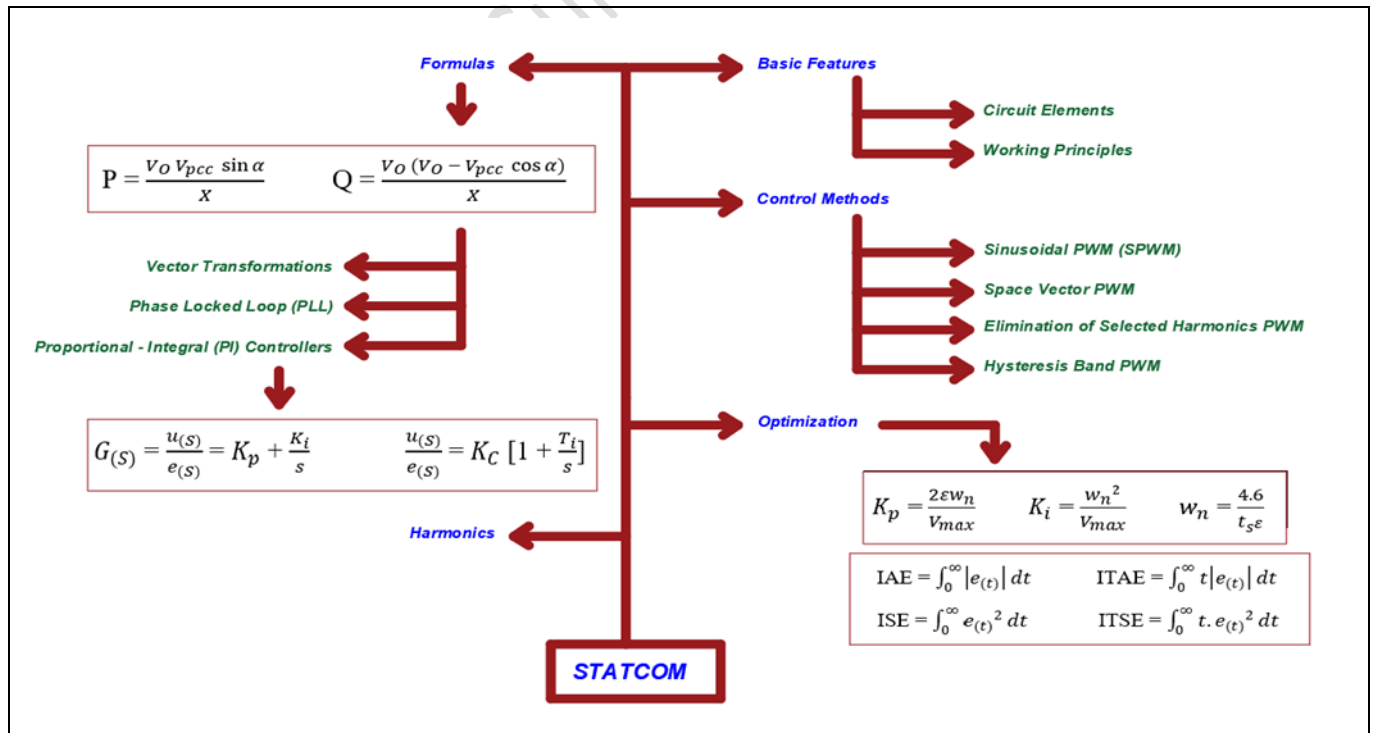


Figure 4. STATCOM formulas and control parameters.

In the formulas shown in Figure 4, the value shown as V_o is the output voltage of the STATCOM, V_{pcc} is the voltage at the PCC point and the angle α is the angular difference between V_o and V_{pcc} . X represents the line reactance.

- P symbolizes active power; Q symbolizes reactive power.
- If V_o is greater than V_{pcc} , the reactive power flow is from V_o to V_{pcc} and the STATCOM is consuming reactive power.
- If V_o is less than V_{pcc} , the reactive power flow is from V_{pcc} to V_o and the STATCOM is producing reactive power.
- $u_{(s)}$: Controller output
- $e_{(s)}$: Error signal
- K_p : Proportional gain, $K_p = \frac{2\varepsilon w_n}{V_{max}}$
- K_i : Integral gain, $K_i = \frac{w_n^2}{V_{max}}$
- T_i : $\frac{K_i}{K_c}$ (Integral time constant)
- w_n : $\frac{4.6}{t_{se}}$ (natural frequency)

The discrepancy between the actual output value and the intended input value is known as the error signal. After the error signal is sent, the PI performs the integral calculation of the error signal. The controller's signal shows that the sum of the integral products of the integral gain and error value equals the gain and error value determined in terms of ratio. A new output signal is sent to the supervised system. The output signal is sent to the sensor, generating a new error signal. When a new error signal arrives, the controller repeats the same steps and so on.

The PI controller is the most often used algorithm for feedback controllers. It is a popular control technique with many uses

nowadays. Systems having nonlinearity, unknown dynamics, and temporal delays are unsuitable for traditional PI controllers. To get an appropriate reaction, PI settings must be automatically adjusted. The time domain criterion is used to evaluate the PI controller in this paper. Longer response times and reduced steady-state errors are the outcomes of good time-domain control parameters like P and I .

The features of the control system may worsen and the system may become unstable if the tuning constants are chosen wrong. Consequently, correctly setting the controller parameters and selecting appropriate tuning constants play an important role in the superior performance of the controller [24].

Integral Absolute Time Error (ITAE) is the most common performance criterion. Figure 4 displays the mathematical expression. The Integral Absolute Error (IAE) and Integral Squared Error (ISE) criterion have the drawback of loading all errors identically, regardless of time, which results in a long reaction time even when the reduction time is quite short. Figure 4 displays the formulas for these requirements. The Integral Squared Time Error (ITSE), whose mathematical expression is displayed in Figure 4, and the ITAE criterion can be used to overcome this shortcoming in the IAE and ISE criteria.

4 Materials, methods and simulation

4.1 Dataset and applied preprocessing

The Kiziltepe district of Mardin province provided the dataset used in the study. The aforementioned territory is situated within the jurisdiction of the 16th TEIAS Regional Directorate, which is headquartered in Batman. Figure 5 shows a single line schematic of the region's electrical infrastructure [29].

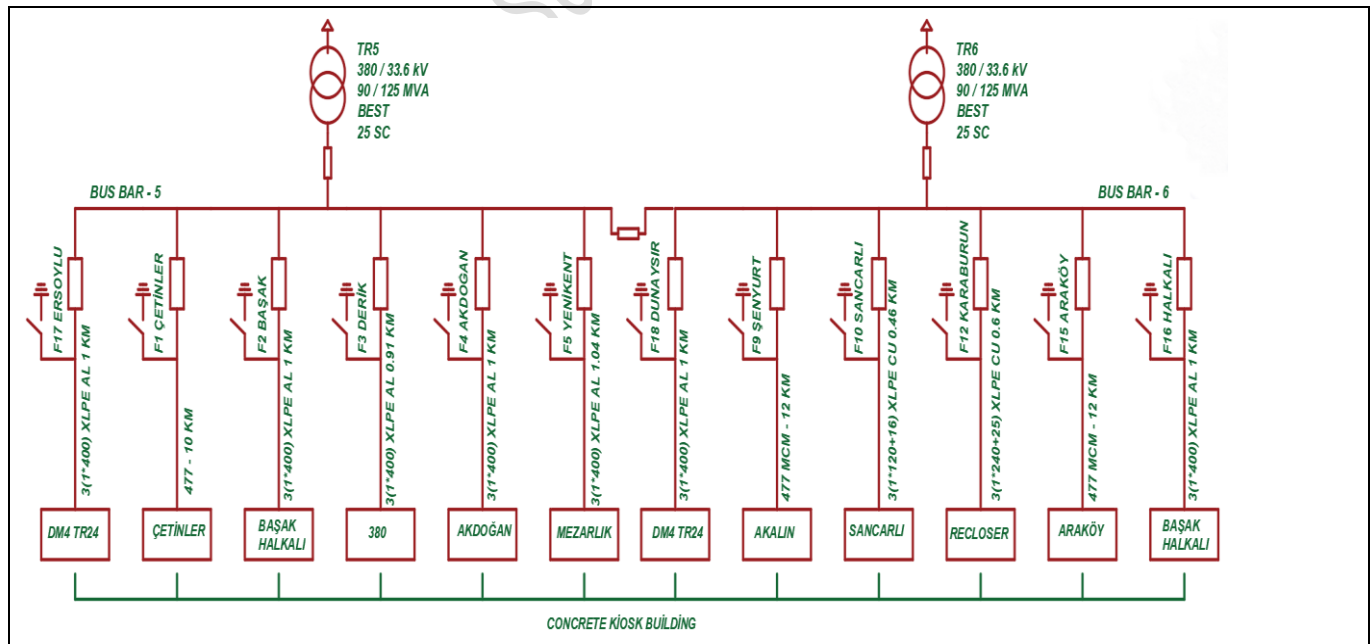


Figure 5. Kiziltepe-2 transformer center single line diagram.

The region in question is fed by 2 main transformer centers named TR5 and TR6. The transformers produce a power of around 90/125 MVA under a voltage of 380/33.6 kV.

4.2 STATCOM data

In the article, since the application area is Türkiye, the project

information is original and taken from the relevant institution, therefore Figure 6 is Turkish data. A 400 kV-2*12 MVar STATCOM circuit was established in the area to maintain voltage stability in the electrical transmission lines, avoid active-reactive power losses, and give customers efficient, high-quality energy [29]. (Figure 6).

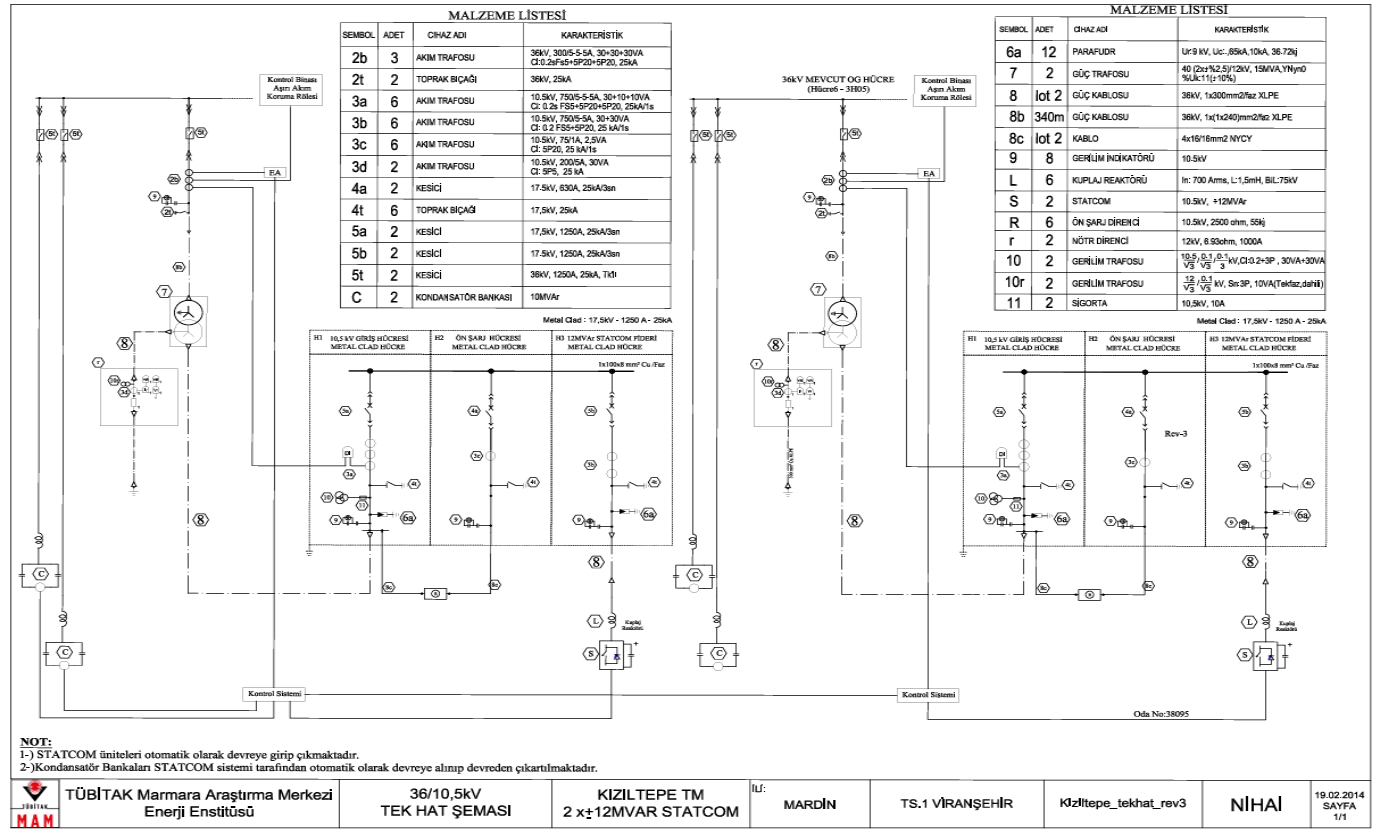


Figure 6. Integration of STATCOM circuit into power system [29].

The power flow data of transformers TR5 and TR6, during the periods when STATCOM was integrated into the power

system, between 01.06.2022 and 30.06.2022 are shown below.

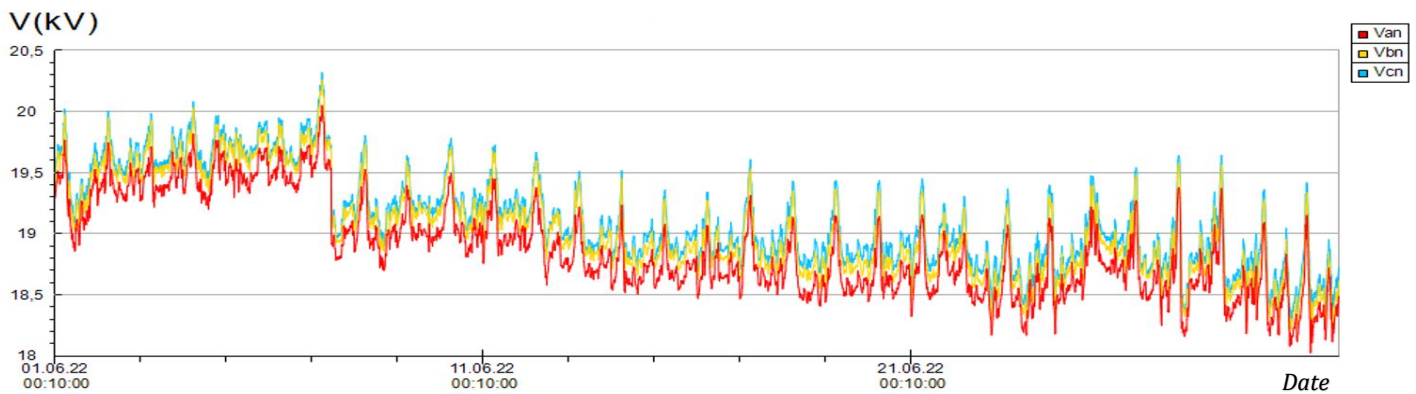


Figure 7. TR5 voltage values (01.06.2022 – 30.06.2022) [29].

The voltage values of the A, B and C phases of the TR5 transformer between 01.06.2022 and 30.06.2022 are shown

in figure 7. It was observed that all 3 phases were loaded at a voltage of 18–20.5

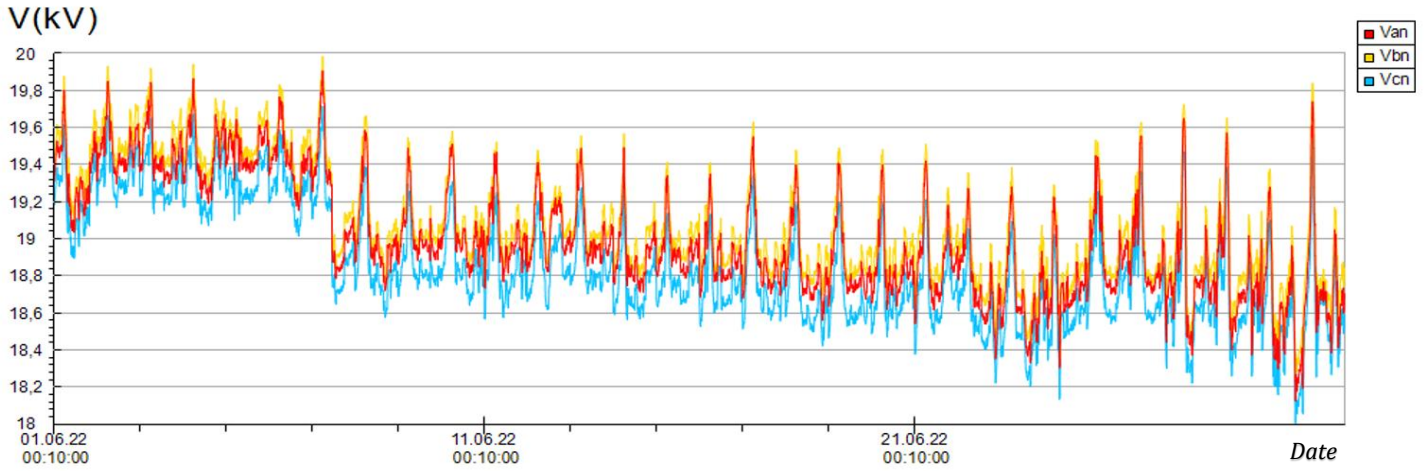
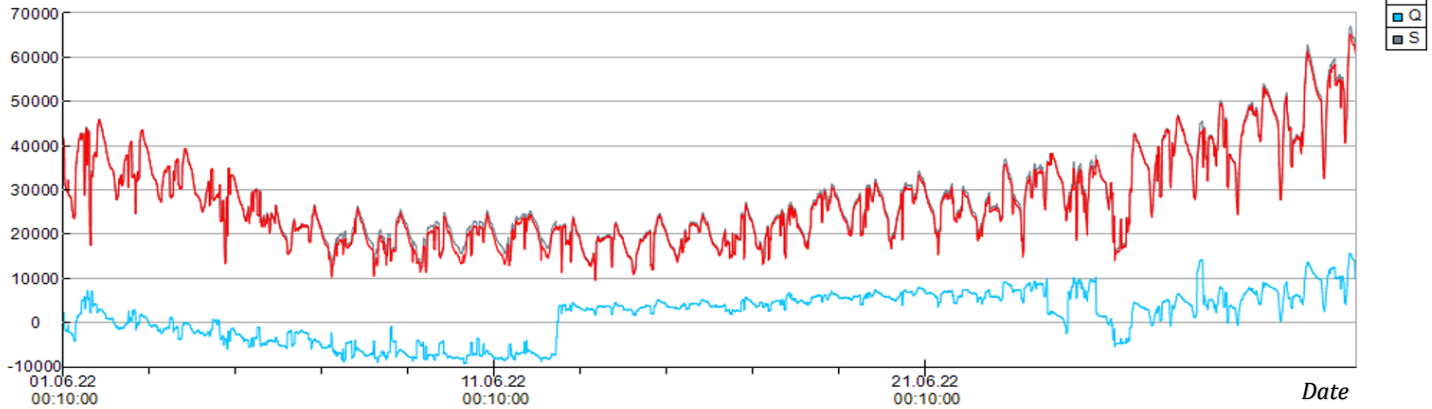


Figure 8. TR6 voltage values (01.06.2022 – 30.06.2022) [29].

Figure 8 displays the voltage values of the TR6 transformer's A, B, and C phases from June 1, 2022, to June 30, 2022. This rate was found to be at 18–19.8 kV levels in phase C, whereas

the voltage values of the A and B phases range from 18.2–20 kV levels.

P (kW), Q (kVar), S (kVA)



P (kW), Q (kVar)

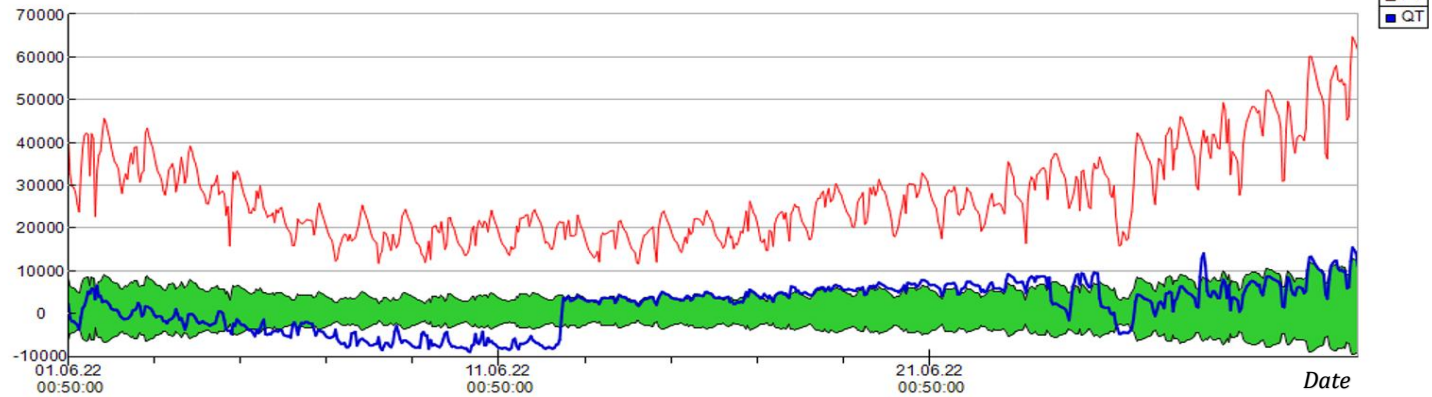
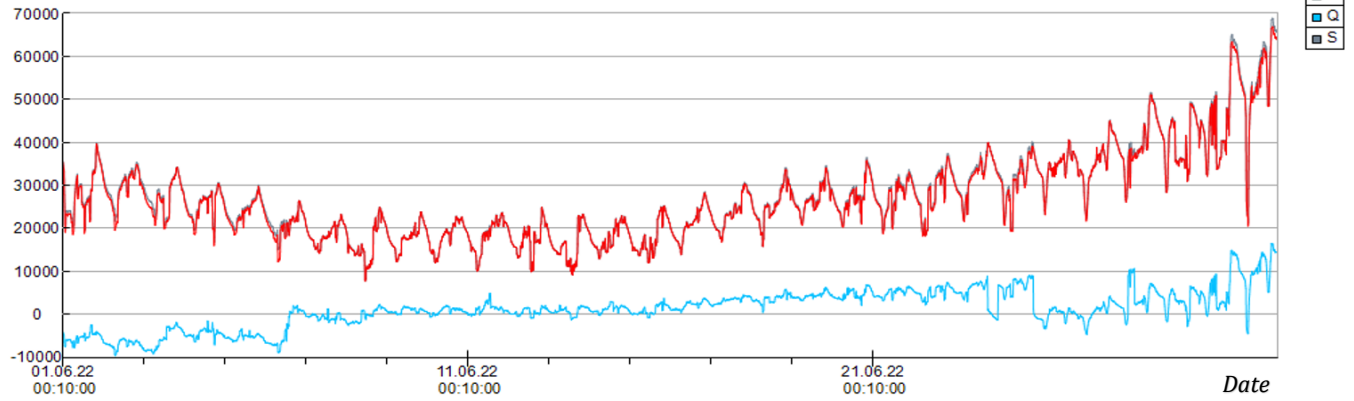


Figure 9. TR5 total power values (01.06.2022 – 30.06.2022) [29].

The total power values of transformer number TR5 between 01.06.2022 and 30.06.2022 are shown in figure 9. It has been observed that the active power values of the system are

around 10-70 MW, apparent power values are around 10-70 kVA, while reactive power values are between -10 and +10 kVar.

P (kW), Q (kVar), S (kVA)



P (kW), Q (kVar)

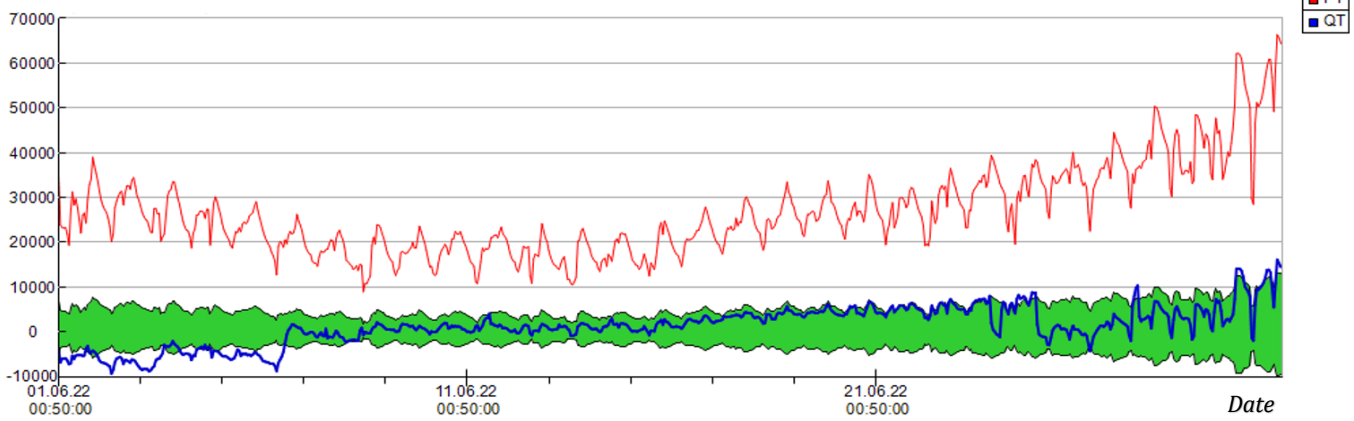


Figure 10. TR6 total power ratings (01.06.2022 – 30.06.2022) [29].

The total power values of transformer number TR6 between 01.06.2022 and 30.06.2022 are shown in figure 10. It has been observed that the active power values of the system are

around 10-70 MW, apparent power values are around 10-70 kVA, while reactive power values are between -10 and +10 kVar.

Value

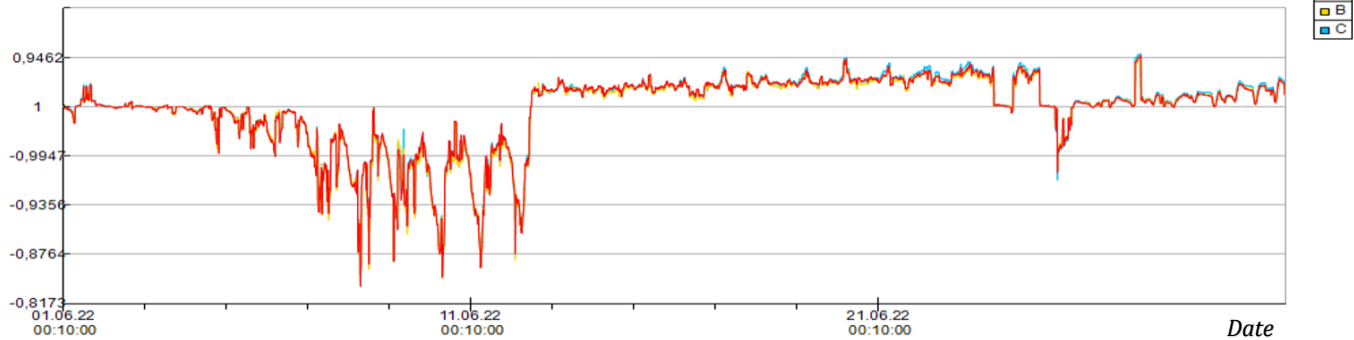


Figure 11. TR5 power factor values (01.06.2022 – 30.06.2022) [29].

The power factor values of the TR5 transformer for phases A, B and C between 01.06.2022 and 30.06.2022 are shown in

figure 11. It is seen that all 3 phases vary between -0.817 and +0.946 levels.

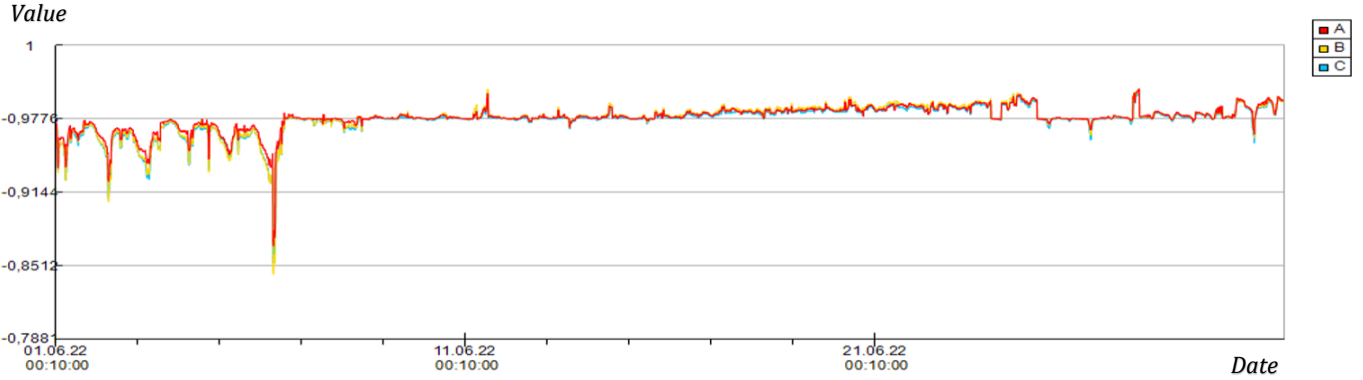


Figure 12. TR6 power factor values (01.06.2022 – 30.06.2022) [29].

The power factor values of phases A, B and C of transformer number TR6 between 01.06.2022 - 30.06.2022 are shown in figure 12. It is seen that all 3 phases vary between -0.851 and +0.978 levels.

With the VSC (Voltage-Sourced Converter) component, the STATCOM control system keeps the phase voltage stable. If the

voltage in the power system is higher or lower than the generated system voltage, the reactor power is either produced or absorbed via the STATCOM. With the integration of the STATCOM circuit into the system, STATCOM data between 01.06.2022 and 30.06.2022 were recorded. The recorded data is shown in the figures below.

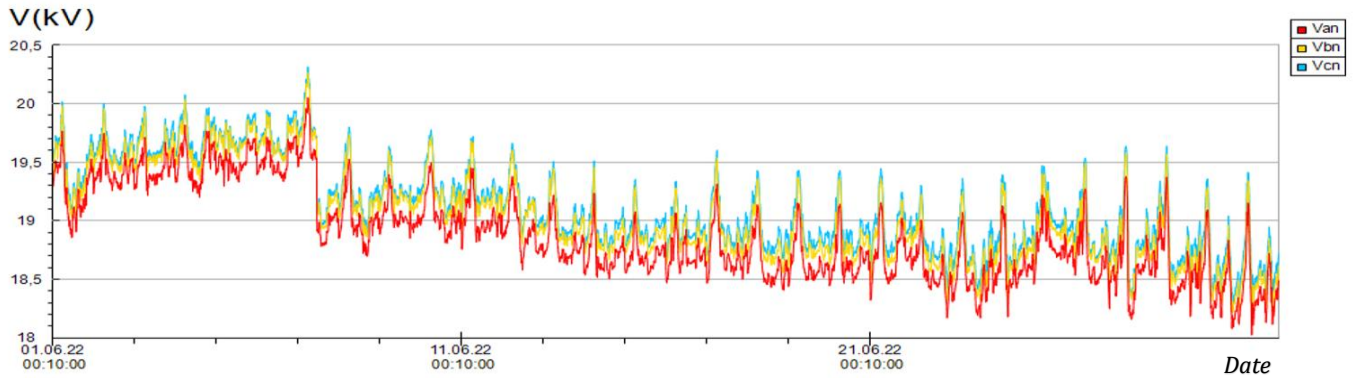
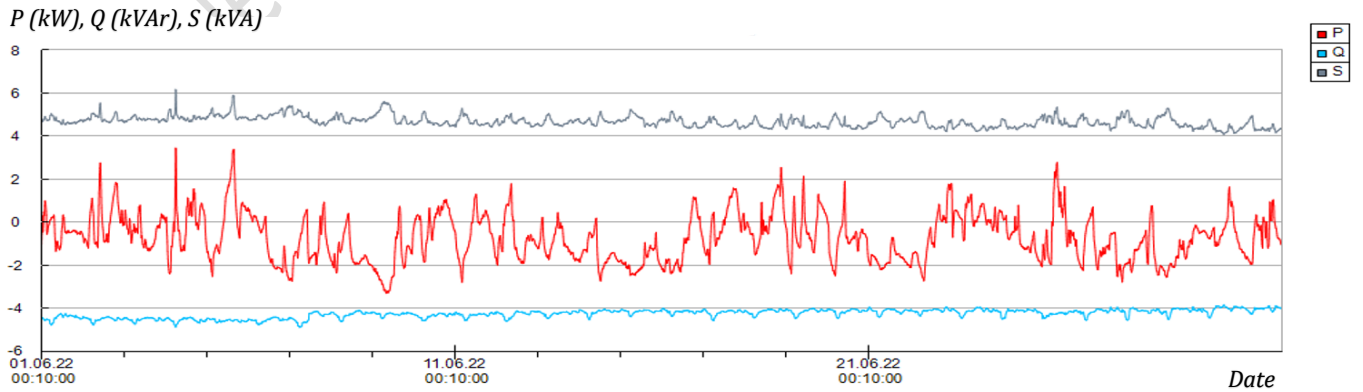


Figure 13. STATCOM voltage values (01.06.2022 – 30.06.2022) [29].

The voltage values of phases A, B and C of the STATCOM circuit between 01.06.2022 and 30.06.2022 are shown in figure 13. It

was observed that all 3 phases were loaded under a voltage of 18–20.5 kV



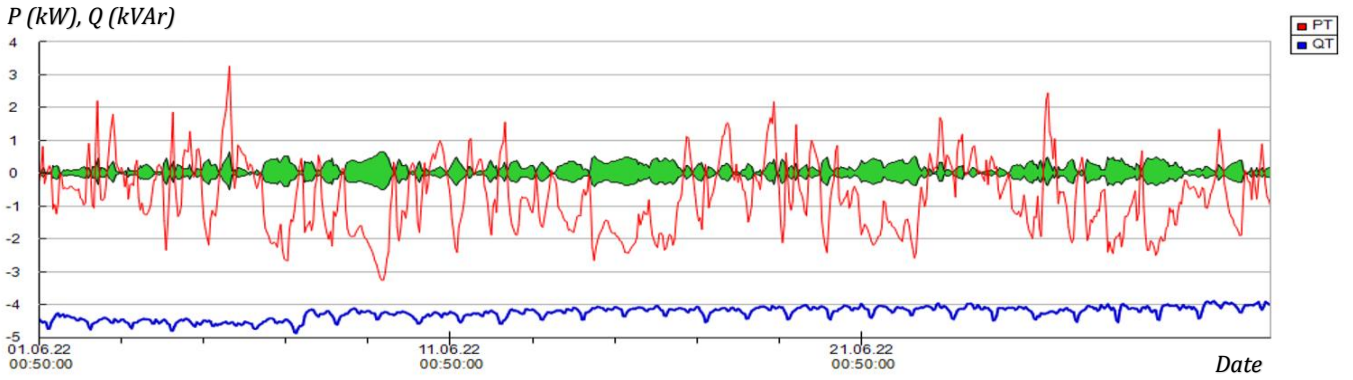


Figure 14. STATCOM total power ratings (01.06.2022 – 30.06.2022) [29].

The total power values of the STATCOM circuit between 01.06.2022 and 30.06.2022 are shown in figure 14. It was observed that the active power values of the system were

between -4 and +4 MW, apparent power values were between 4 and 6 kVA, while reactive power values were between -4 and -5 kVar.

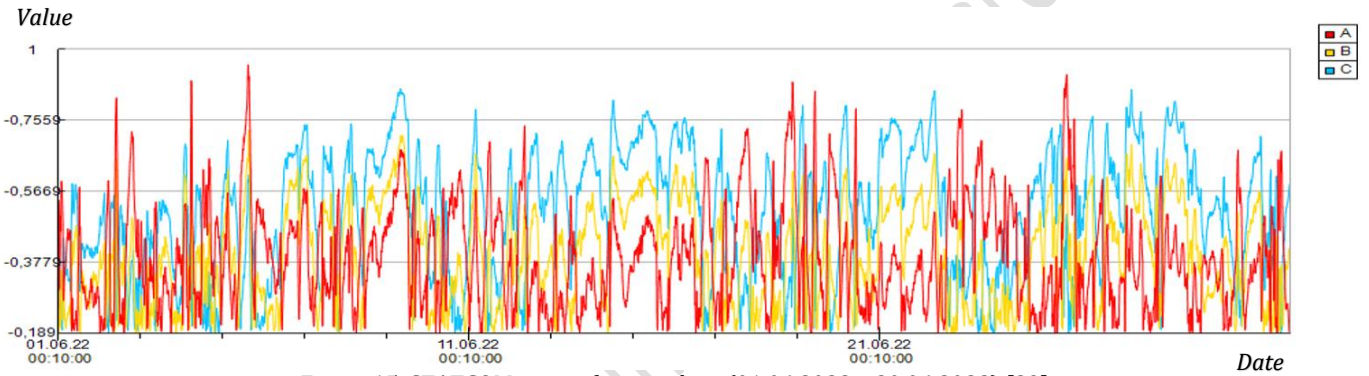


Figure 15. STATCOM power factor values (01.06.2022 – 30.06.2022) [29].

The power factor values of the A, B and C phases of the STATCOM circuit between 01.06.2022 - 30.06.2022 are shown in figure 15. Accordingly, the power factor varies between -0.189 and -0.756 in phases A and C, while this ratio is between -0.189 and -0.567 in phase B.

4.3 Modeling and simulation of power system on MATLAB / Simulink program

Figures 4-14 display actual statistics from the TEİAŞ 16th Regional Directorate. The single line diagram of the area and the circuit structure of the STATCOM device integrated into the area under the direction of TÜBİTAK are displayed in Figures 5 and 6 if examined actual data collected from the field. Figures 6, 7, 8, 9, 10 and 11 show the actual voltage values, actual power values and actual power factor values of the power system in the field. Figures 12, 13 and 14 show the actual voltage values, actual power values and actual power factor values of the STATCOM device installed in the field. When figures 6, 7 and 12 are evaluated together, it shows that the voltage values on the STATCOM device are in line with the voltage values of the main transformers 5 and 6 that feed the power system in the field. Figures 10 and 11 show that there were fluctuations in power factor values starting from 01.06.2022 and these fluctuations continued until 30.06.2022. When these two situations are evaluated together, it is understood that the STATCOM device was activated to bring these power factor values, which fluctuated between 01.06.2022 and 30.06.2022, back to standard values, as can be

seen in Figure 14. Considering the data in Figure 13, it shows the amount of power consumed by the STATCOM device in order to keep the power factor values in the field at normal levels. These actual data, particularly those from figures 12, 13, and 14, demonstrate that the region's power supply had significant voltage variations and that the STATCOM device was in operation between June 1, 2022, and June 30, 2022, in order to counteract these voltage sags. Similarly, an analysis of Figures 8 and 9 reveals that each of the primary supply transformers, TR5 and TR6, generates an average load of around 30 MW, resulting in an average total load of about 60 MW in the field.

The Matlab/Simulink program was used to begin system modeling in light of the collected real data. The system was modeled using entirely genuine data. First, the primary supply transformers TR5 and TR6 with 33.6 kV voltage values, 125 MVA power, and an average load of 60 MW power were included to the model based on the actual data collected from figures 4–14 in the field. The TEİAŞ 16th Regional Directorate was asked to provide the precise lengths of the 12 feeder lines in the area, which were then entered into the model's system. The simulation phase started once the $2 \times 12 = 24$ MVar STATCOM device installed into the region by TÜBİTAK was added to the program's model.

A similar power system in the region in question has been modeled on the Matlab/Simulink program as seen in figure 16.

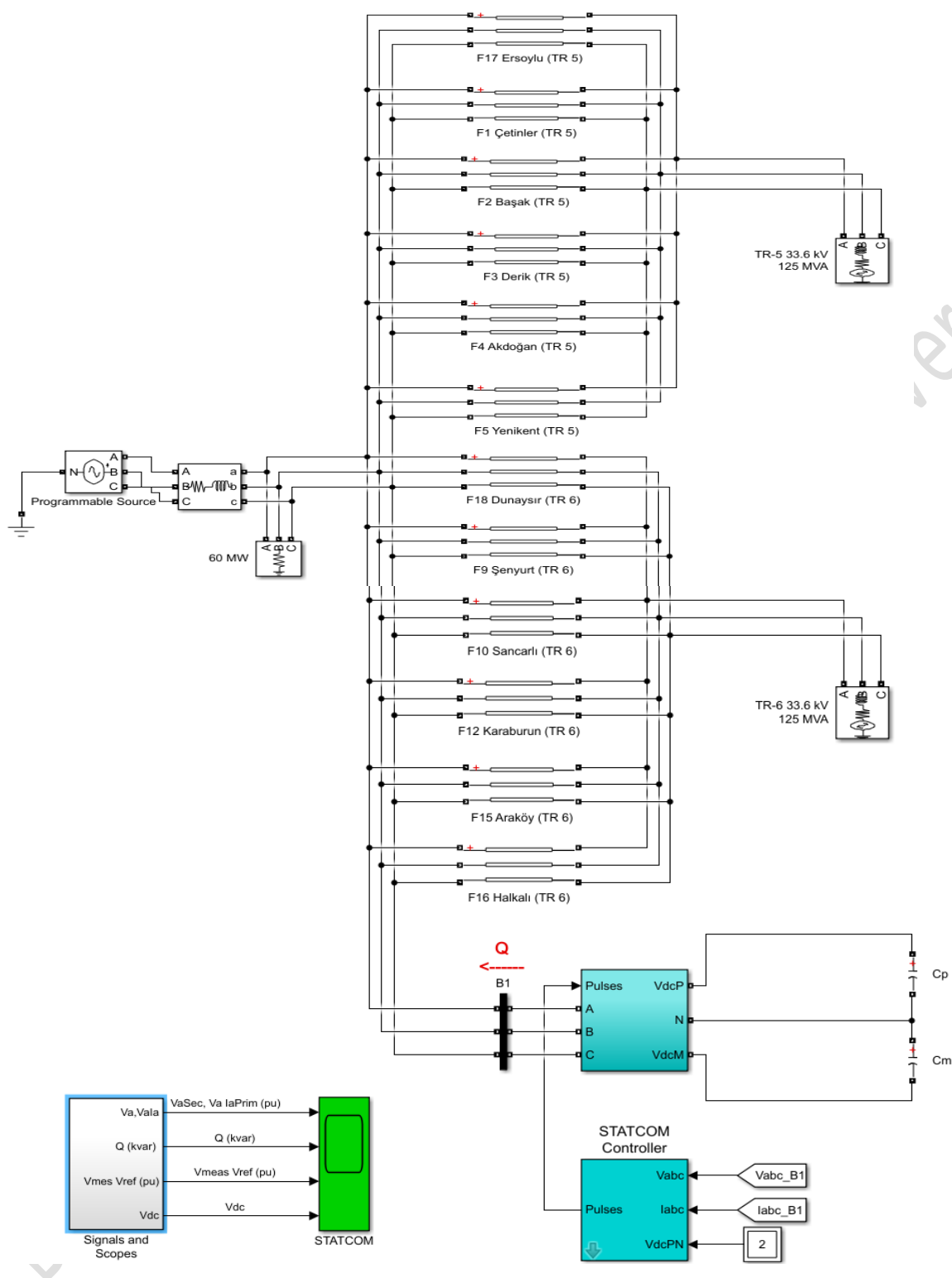


Figure 16. Modeling of the power system in Matlab/Simulink program.

The simulation was started by modeling a similar version of the power system on the Matlab / Simulink program and the

Phase Voltage Values

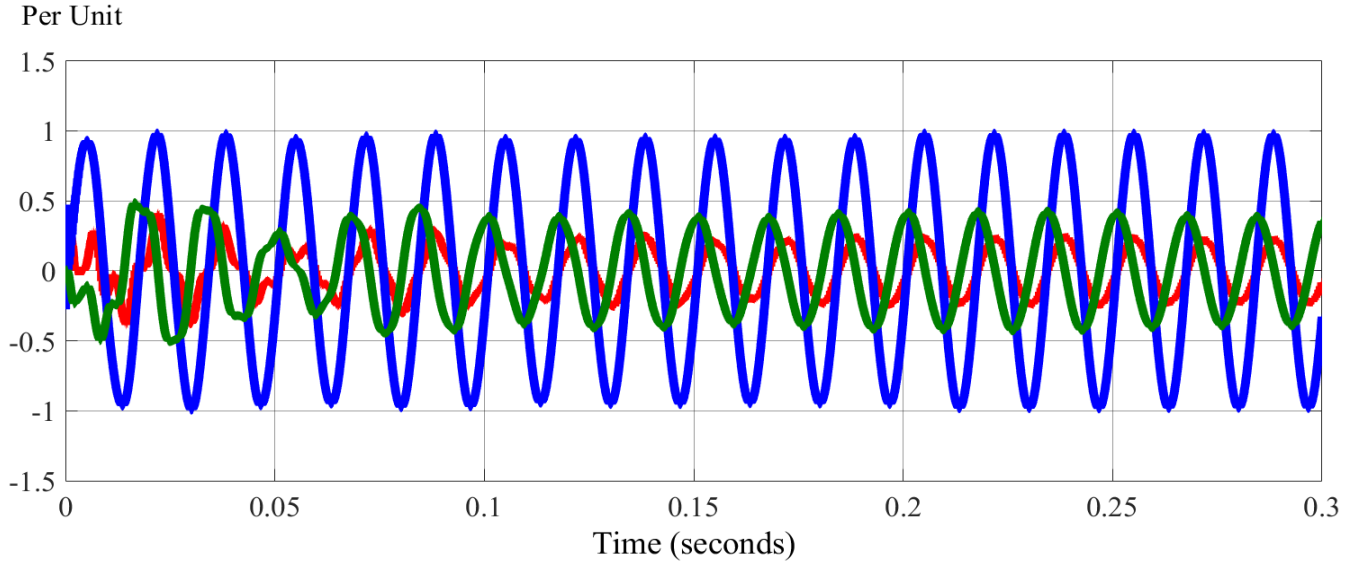


Figure 17. Phase voltage values resulting from simulation.

Phases 1, 2 and 3 in figure 17 are represented by the colors red (phase A), dark blue (phase B), and dark green (phase C) respectively. Voltage collapses happened in phases A and C due to significant voltage swings, even though no degradation is seen in phase B.

Phase A, which is represented in red; and phase C, which is represented in dark green, both undergo voltage collapses between 0 and 0.155 seconds, according to figure 17. As illustrated in figure 18, the STATCOM responds rapidly when it senses these voltage falls in the phases.

Reactive Power Value

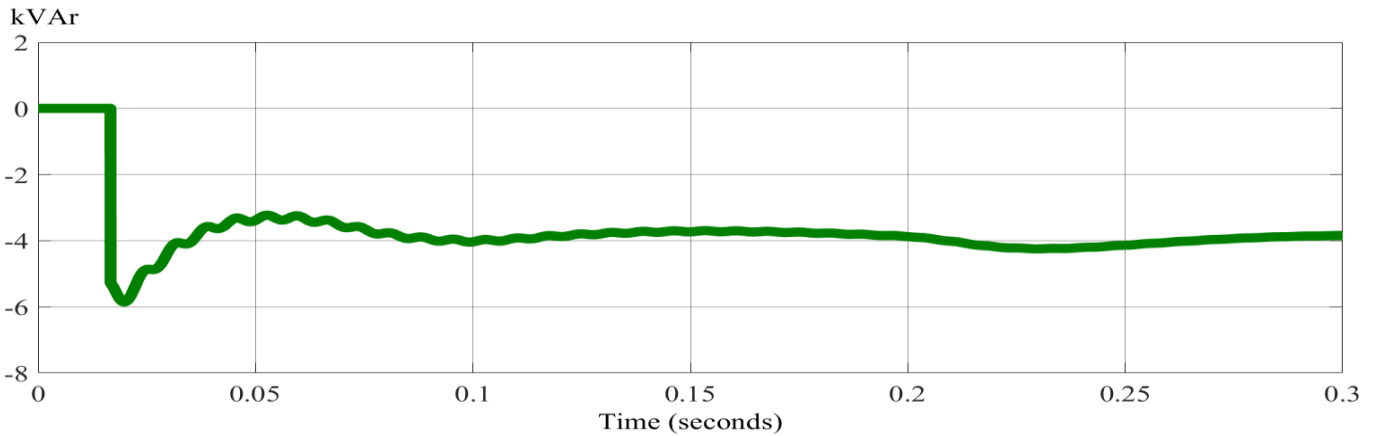
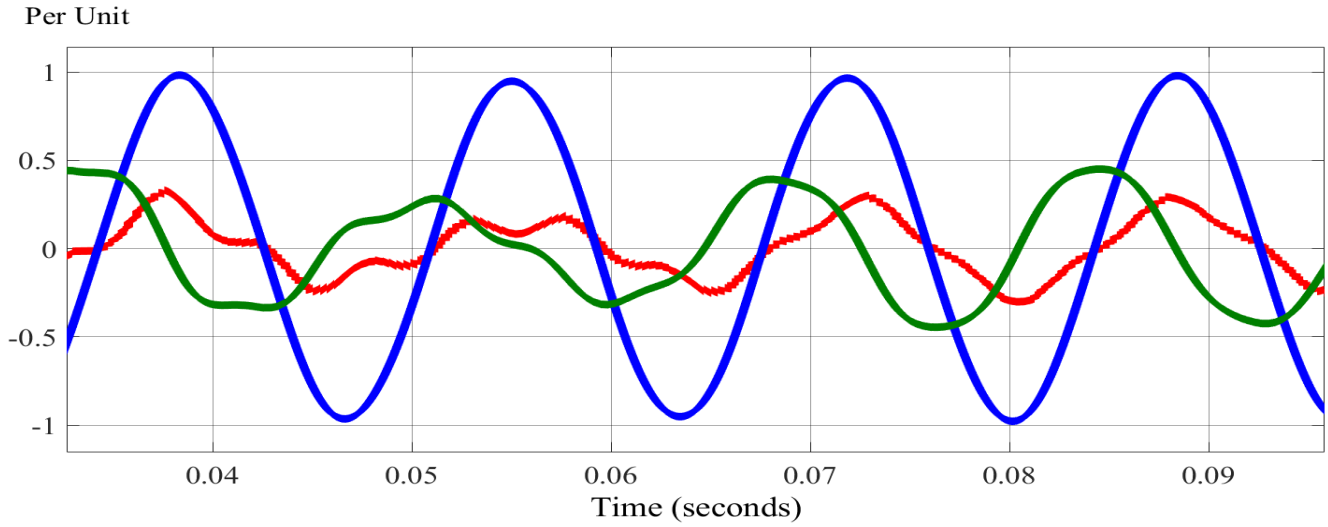


Figure 18. Reactive power values resulting from simulation.

STATCOM swiftly activates when it detects voltage falls in the power supply, as seen in figure 18. Within 0.01 to 0.02 seconds, or almost a hundredth of a second, it makes an intervention in the system's reactive power. With the intervention of STATCOM to the system, the voltage fluctuation in phase 3

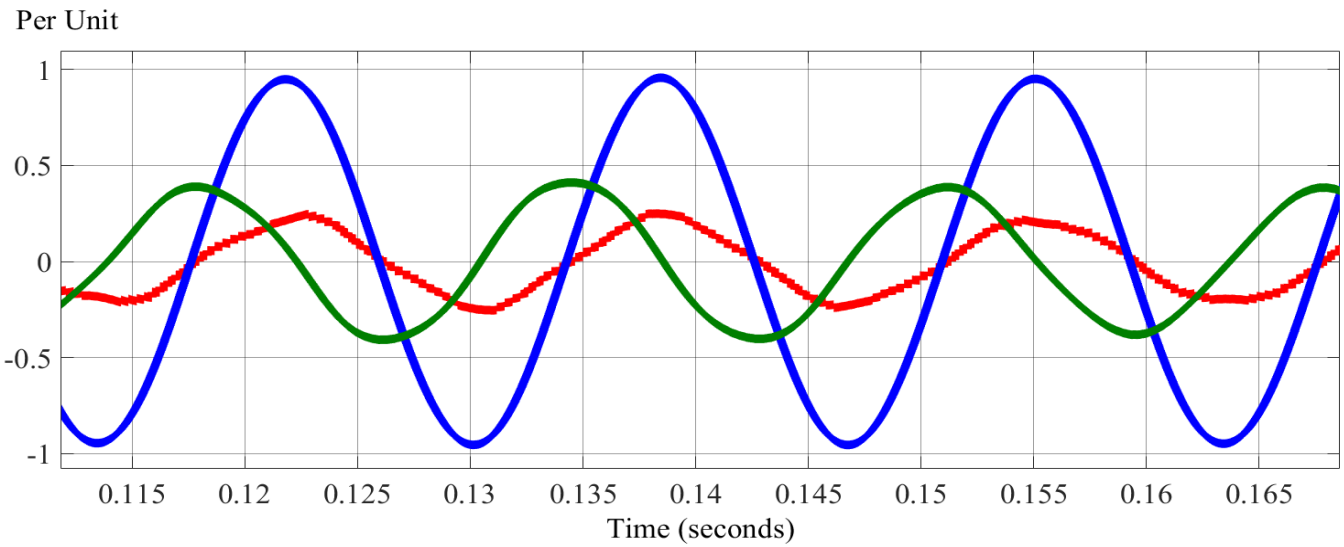
(shown in dark green) is 0.07. while it recovers in seconds, the voltage fluctuation in phase 1 (shown in red) is 0.155. it has been corrected in seconds. Thus, the voltage sags occurring in the power system were eliminated within 0.155 seconds with the effect of STATCOM. (Figure 19)

Phase Voltage Values



(a)

Phase Voltage Values



(b)

Figure 19. Phase voltage values: (a) Voltage recovery time of phase 1, (b) Voltage recovery time of phase 3.

The STATCOM device gives the power system an actual reactive power of about -4 kVAr, as seen in Figure 13. The simulation's output in Figure 18 shows that the model's STATCOM device gives the system a reactive power of about -4 kVAr. This made it possible for us to verify that the model we had developed was accurate and that the simulated results agreed with the real data.

In figure 20, the amount of reactive power obtained as a result of the simulation is compared with the actual data. When the actual reactive power data obtained from the STATCOM device between 01.06.2022 and 30.06.2022 in the region in question is examined, it is observed that the reactive power values obtained from the STATCOM device as a result of the simulation are almost the same.

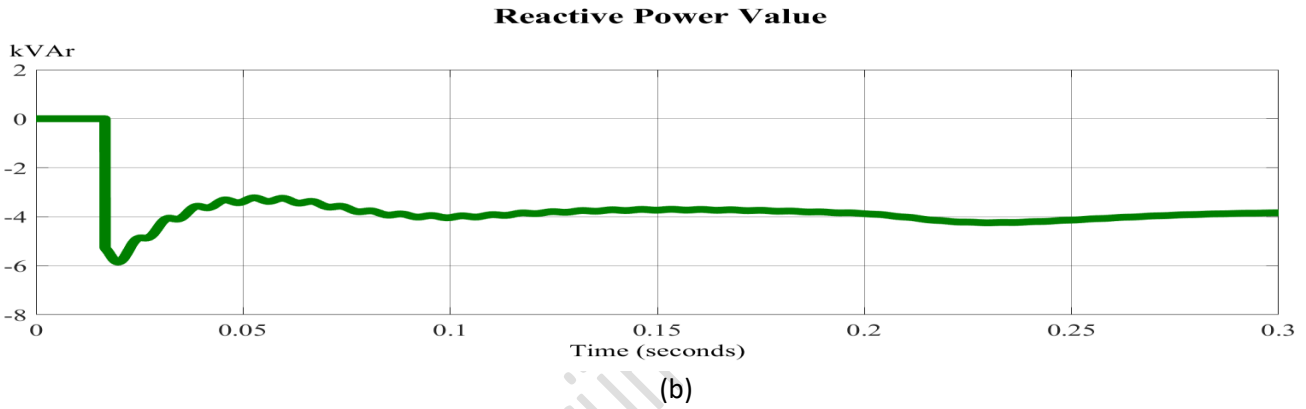
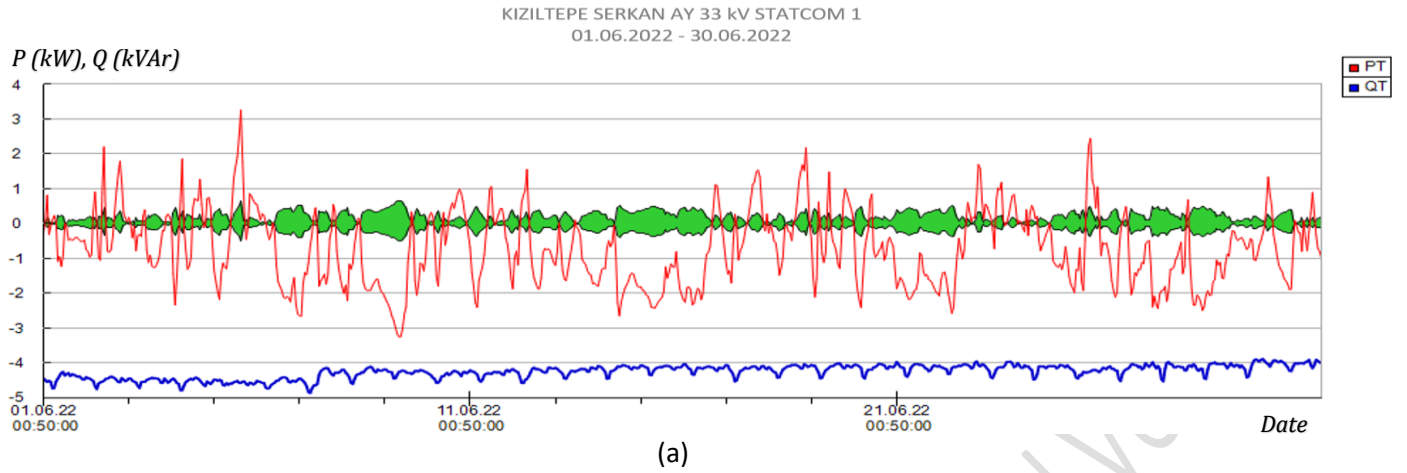


Figure 20. Power ratings: (a) Actual power values at the application site, (b) Reactive power value resulting from simulation.

The voltage collapse in the power system at 0.01 seconds caused STATCOM to be activated, as can be seen when looking at figure 21. The graph shows the changes in the system's voltage values in terms of units. The system's voltage value was 1 (one) in per unit until 0.01 seconds, at which point it fell

to 0.95 levels as a result of voltage oscillations in the phases. But after STATCOM entered the system, this value swiftly rose to 0.99 levels, and the system's power flow achieved an effective and high-quality level.

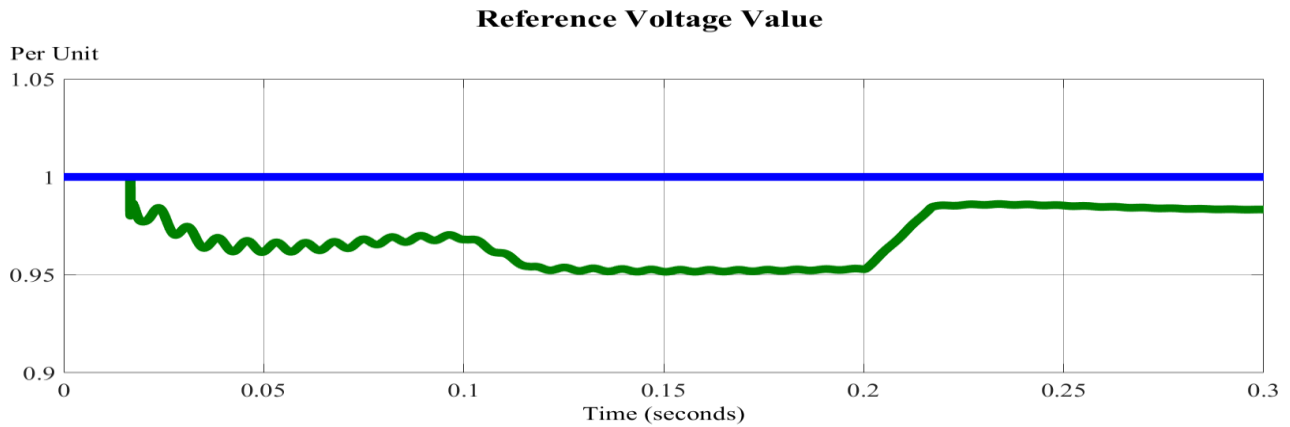


Figure 21. Voltage recovery time according to reference value.

5 Conclusion

This article explains how voltage was controlled using the STATCOM device, which is a component of the energy transmission system's distribution busbar in a 400 kV and 2*12 MVar power system in the Kiziltepe District of Mardin Province and is overseen by the TEIAS 16th Regional Directorate.

It is evident by analyzing the graphs of simulation results and actual data that STATCOM is a significant, potent, and extremely quick balancing factor on the energy transmission system. Consumers benefit from a higher quality and more efficient energy flow since the negative forces that are created on the system rapidly become passive and the voltage value variations are balanced.

It is clearly seen that the STATCOM device provides a rapid compensation in terms of time compared to the mechanical switching devices that have been in use in the past. It has been observed that the STATCOM device rapidly performs the compensation required by the system as a result of the disturbances that may occur on the power system due to inductive and capacitive loads. It has been observed that the dynamic response of the compensator is fast and that it protects the power system on which it is located by balancing it despite sudden load changes and different load characteristics.

The results obtained from the simulation showed that the reactive power that may occur on the power system has negative consequences in terms of voltage stability and values of the power system. The power system's bus voltage value is brought to the required reference value by STATCOM, which is engaged quickly to counteract voltage sags, which are the most significant adverse impacts. It has been demonstrated that STATCOM helps to boost the transmission system's capacity and dependability while also improving its voltage stability.

Ultimately, STATCOM can play a significant role in not only reactive power and voltage regulation but also in detecting human-induced electricity theft during the electricity distribution process. It is possible to ascertain if a voltage breakdown is caused by humans or by nature by surveying the area where STATCOM finds one. By preventing unlawful power use in the area, STATCOM can improve the efficiency and quality of energy use in the area. Additionally, defects that could potentially create disruptions in the region's energy supply can be identified and avoided beforehand by analyzing the voltage changes in the electrical distribution areas identified by STATCOM.

6 Author contribution statement

In the study carried out, **author 2** contributed to the formation of the idea, design and literature review; **author 1** contributed to the evaluation of the obtained results, procurement of the materials used, examination of the results and checking the content of the article.

7 Ethics committee approval and conflict of interest declaration

There is no need to obtain ethics committee approval for the prepared article.

There is no conflict of interest with any person/institution in the prepared article.

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